MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON
1/ TYPE DOD-035

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Coaxial, fixed frequency, 9,355 to 9,395 MHz, integral magnet, air cooled, 40-kw minimum peak power output

ABSOLUTE RATINGS:

Parameter:	Ef	If	if(surge)	tk	tpc	prr	Du	rrv	ib
Unit:	V	A	a	sec	µs	pps	---	kv/µs	a
Maximum:	14	2.6	10	---	1.5	2,500	0.001	160	20
Minimum:	---	---	---	120	0.05	---	---	90	7
	(Note 1)				(Note 2)			(Note 3)	

ABSOLUTE RATINGS:

Parameter:	Pi	pi	Temperature		Pressurization
			Body	Input bushing	Input and output
Unit:	W	kw	°C	°C	psia
Maximum:	160	250	+150	+165	45
Minimum:	---	---	-55	-55	10
			(Note 4)	(Note 4)	(Note 5)

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Mounting support: Flange (see figure 1)
Input coupling: See figure 1; notes 19 and 20
Output coupling: See figure 1 (note m); notes 19 and 20
Marking: See figure 1 (see note h)
Cooling: Forced air (see note 6)
Handling: See note 7
Weight: 7.5 pounds (maximum)

TEST CONDITIONS:

Test condition (1):

Parameter:	Ef	tk	tpc	Du	rrv	epy	ib	VSWR	TA
Unit:	V	sec	µs	---	kv/µs	kv	a	---	°C
Maximum:	11.5	---	0.11	---	130	13	12.5	1.10	100
Nominal:	11.1	---	0.10	0.000225	100	12	12.0	---	---
Minimum:	10.6	90	0.095	---	90	11	11.5	---	---
	(Note 1)		(Note 2)		(Note 3)				

Test condition (2):

Parameter:	Ef	tk	tpc	Du	rrv	epy	ib	VSWR	TA
Unit:	V	sec	µs	---	kv/µs	kv	a	---	°C
Maximum:	11.2	---	0.55	---	130	13	12.5	1.10	100
Nominal:	10.7	---	0.50	0.000375	100	12	12.0	---	---
Minimum:	10.1	90	0.45	---	90	11	11.5	---	---
	(Note 1)		(Note 2)		(Note 3)				

GENERAL:

Qualification - Required

1/ Replaces:
Litton tube type L - 4624
Raytheon tube type QKH-1820
Varian tube type VMX-1146
Ridge Electronics Corporation Drawing No. OA1100002

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1032	Vibration, mechanical	8,9,10	2		MP FM	--- ---	0.5 1.0	% MHz
1041	Shock, specified pulse	11	-	Test condition K; no voltages applied	---	---	---	---
1047	Low-temperature operation	4,10,12	2	TA = -55°C	MP	---	0.5	%
4027	Temperature coefficient	4,13	2	TE = 50° to 150°C	ΔF/ΔT	---	0.18	MHz/°C
4030	Thermal frequency drift	4,14	2	t = 20 minutes	ΔF	---	10	MHz
---	Frequency modulation	15	2		FM	---	1.0	MHz
	Quality conformance inspection, part 1	16	-					
4003	Pressurizing	17,18	-	45 psia (min) input and output	---	---	---	---
4289	Heat current, nonoperating	-	-	Ef = 12.6 V; tk = 120 (max)	If	2.0	2.4	A
---	Coupling	19,20	-	VSWR = 1.1:1 (max), except as noted	---	---	---	---
4303	Warmup time	1	-	Ef = 12.6 V; tk = 120 (max)	---	---	---	---
Ⓐ4250	Power output (1)	-	1		Po	9	---	W
Ⓐ4250	Power output (2)	-	2		Po	15	---	W
4306	Pulse voltage	-	1, 2		epy	11	13	kv
4308	RF bandwidth	21	1, 2		BW	---	2.0/tpc	MHz
4308	Minor lobe ratio	21	1, 2		SL	-10	---	dB
4315	Stability	10	1, 2		MP	---	0.1	%
4315	Starting stability	10, 22	1, 2		MP	---	0.25	%
Ⓐ---	Frequency	23	1, 2		F	9, 355	9, 395	MHz
	Quality conformance inspection, part 2							
1143	Forced cooling	4,6	2		ΔT	---	80	°C
4266	Direct-interelectrode capacitance	-	-	Cathode to mounting plate	Cin	9	14	pF
4310	Frequency pulling figure	24	2		ΔF	---	4	MHz
4311	Frequency pushing figure	25	2	ib = 10 to 15 a	ΔF/Δib	---	0.09	MHz/a
4318	Time jitter	26	2		tj(σ)	---	10	ns
	Quality conformance inspection, part 3							
4551	Cycled life	27,28	-	Group D; VSWR = 1.5 (min), cycled through λg every 15 minutes (approx)	t ---	--- ---	1,250 1,500	hrs Cycles

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued							
---	Cycled life-test end points							
(A) ---	Frequency	23	2		F	9,355	9,395	MHz
(A) 4250	Power output (2)	-	2		Po	12	---	W
4308	RF bandwidth	21	1, 2		BW	---	3.0/tpc	MHz
4308	Minor lobe ratio	21	1, 2		SL	-6	---	dB
4315	Stability	10	2		MP	---	0.5	%

NOTES:

1. Prior to the application of high voltage, the cathode shall be heated to the required initial operating temperature by applying 12.6 volts ±5 percent for 120 seconds minimum. Upon the application of anode voltage, the heater voltage shall be reduced according to the following schedule:

	Du	Ib (mAdc)	Ef(V) ±5%
Standby:	---	---	12.6
Operate:	0.001	15	7.9
For all other conditions:			

$$Ef = 12.6 \left(1 - \frac{Pi}{600}\right) \pm 10\%$$

2. The characteristics of the applied pulse shall be those which result in proper starting and oscillation. The rate of rise of the voltage pulse, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations.

3. The rate of rise voltage (rrv) shall be measured in accordance with method 4304, and shall be defined by the steepest tangent to the leading edge of the voltage pulse above the 70 percent amplitude point. Any capacitance used for measuring shall not exceed 6 pF.

4. The temperature shall be measured at the point indicated on figure 1.

5. The TUT shall be capable of normal operation without electrical breakdown with the input bushing in air at normal atmospheric conditions.

6. An airflow of approximately 15 cfm at approximately 760 mmHg at 25°C, shall be directed toward the body of the TUT from an orifice of 2-1/2 by 1-3/16 inches placed 1/2 to 3/4 inch from the body. The specified rise above ambient temperature shall not be exceeded.

7. Care shall be excercised when handling and mounting the tube to prevent demagnetization. The use of magnetic inspection shall be prohibited.

8. The TUT shall be mounted in a resonance free jig and vibrated with sinusoidal excitation in each of three mutually perpendicular planes through the following amplitudes:

15 to 44 Hz	10 G (or 0.1 inch DA max)
45 to 100 Hz	10 G

Cycling test: The frequency shall vary from 5 to 500 to 5 Hz with approximately logarithmic progression, and shall require approximately 30 minutes to traverse the range. This constitutes one cycle. The TUT shall be vibrated for one such cycle in each of the three planes.

Resonance test: Mechanical resonant frequencies of the tube shall be determined during the cycling test.

Vibration test schedule (times shown refer to one axis)

Type	Room temperature
Resonance	30 minutes
Cycling	30 minutes

9. At the conclusion of this test, the TUT shall satisfy the requirements of the frequency modulation and stability tests listed herein.

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NOTES: -Continued

10. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percentage of the number of input pulses applied during the period of observation. The missing pulses (MP) due to any causes, are considered to be missing if the rf energy is less than 70 percent of the normal energy level. The stability shall be measured when VSWR of 1.5:1 minimum is introduced in the load and the phase is adjusted at the start of each measurement interval to produce maximum instability. The missing pulse count shall be performed over a 3-minute test interval of a 6-minute test period.

11. The TUT shall be subjected to five shocks of the specified peak amplitude and duration in each of three mutually perpendicular directions. Following impact tests, the tube shall show no mechanical failure and shall meet the power output and pulse voltage requirements of quality conformance inspection, part 1.

12. The TUT shall be stabilized for 4 hours minimum by supplying cooling air at -55°C at 40 cfm maximum. Upon stabilization, a heater voltage of 12.6 V shall be applied for 120 seconds maximum prior to the application of anode voltage. This stability test shall be 3 minute duration to begin after 30 seconds of anode operation.

13. Temperature measurements shall be made only after thermal equilibrium has been reached. The frequency shall be measured at the extremes of any 30°C temperature difference in the specified temperature range.

14. The TUT shall be operated as specified such that the measured temperature shall be between 80° and 100°C. After thermal equilibrium is reached, measure and record the frequency. With no voltages applied, the TUT shall be cooled for 15 minutes minimum prior to the following test:

    The specified anode voltage shall be applied after 120 seconds maximum heater warmup time, then measure the frequency at 3-minute (approximately) intervals until 20 minutes has elapsed. A graph of frequency versus time shall be constructed using this data. The maximum frequency deviation from the previously recorded thermal equilibrium frequency shall not exceed the specified limits. The first reading shall be taken 12 seconds maximum after the application of high voltage.

15. Frequency modulation shall be defined as interpulse frequency changes (clatter). For frequency modulation measurements, the maximum peak-to-peak frequency deviation shall not exceed the specified limits. This test shall be performed using the specified heater voltage at 60 to 400 Hz, and shall exclude thermal drift and pushing effects. The test equipment shall have a bandpass of 10 MHz minimum at the 3 dB points.

16. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

17. The seal formed by clamping the TUT mounting plate against a suitable tube test fixture shall be hermetically tight for 1-minute minimum with the specified air pressure applied so as to surround the entire input bushing below the mounting plate.

18. The seal formed by clamping the TUT output flange against a suitable tube test fixture shall be hermetically tight for 1-minute minimum with the specified air pressure applied internally to the test fixture.

19. The minimum air pressure to assure prevention of electrical breakdown in the output coupling shall be 15 psig for voltage standing wave ratios up to 1.5:1 with phase shift variable over 360 electrical degrees. The TUT shall be coupled directly to a UG-52/U choke flange modified so that mounting holes provide clearance for No. 8 bolts.

20. The modulator shall be such that energy per pulse delivered to the tube, if arcing occurs, shall not greatly exceed the normal energy per pulse. The tube heater shall be protected against arcing by use of a connector that places a minimum of 4,000 pF across the heater directly at the input terminals.

21. The radio frequency bandwidth and side lobes shall be within the limits specified when a VSWR of 1.5 minimum is introduced in the load at a distance of 1 ±0.05 meter from the TUT flange and the phase is adjusted at the start of each measurement to produce maximum degradation. A satisfactory spectrum is one whose slope does not change sign more than once for power levels greater than the specified dB below its peak.

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NOTES: -Continued

22. This test shall be conducted at the conclusion of the holding period. After the holding period, anode voltage shall be applied for 120 seconds minimum after the application of heater voltage. The missing pulse count test interval of 3 minutes shall start immediately after the application of anode voltage. All other provisions of note 10 shall apply.

23. The TUT shall be at thermal equilibrium with cooling air supplied at approximately 15 cfm at an ambient temperature between 20° and 30°C.

24. The frequency deviation (maximum frequency minus minimum frequency) shall not exceed the specified limits when a VSWR of 1.5:1 minimum is varied throughout all phases when introduced into the load at a distance of 1 ±0.05 meters from the TUT flange.

25. The pushing figure shall be measured in 5 ampere step (10 to 15 a), and shall not exceed the specified limits. The peak current through the TUT shall alternately be the limits as specified under this test condition. These tests shall be conducted in a manner to exclude thermal drift effects and frequency instability not due to pushing.

26. Jitter is defined by the rms value σ. Jitter may be measured directly using a L.F.E. Model 5015A Time Jitter Tester, or equivalent and a Hewlett-Packard Model 3400A Voltmeter, or equivalent. Jitter may be viewed on an oscilloscope (50 MHz bandwidth minimum) and measured at the 5 σ limit which contains 99 percent of the pulses.

27. Life test shall be conducted in accordance with the following cycle:

Condition	ib(A)	Ef(V)	Duration (minutes)
Standby	0	12.6	2.0
Test condition 1	12.0	11.1	33.0
Test condition 2	12.0	10.7	15.0
Off	0	0	10.0

28. If during life test, the tube does not meet the specified limits, it shall be recycled for an additional 5 cycles. At such time, the tests shall be repeated. If the tube fails the second test, it shall be considered unsatisfactory.

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

Agent:
DSA - ES

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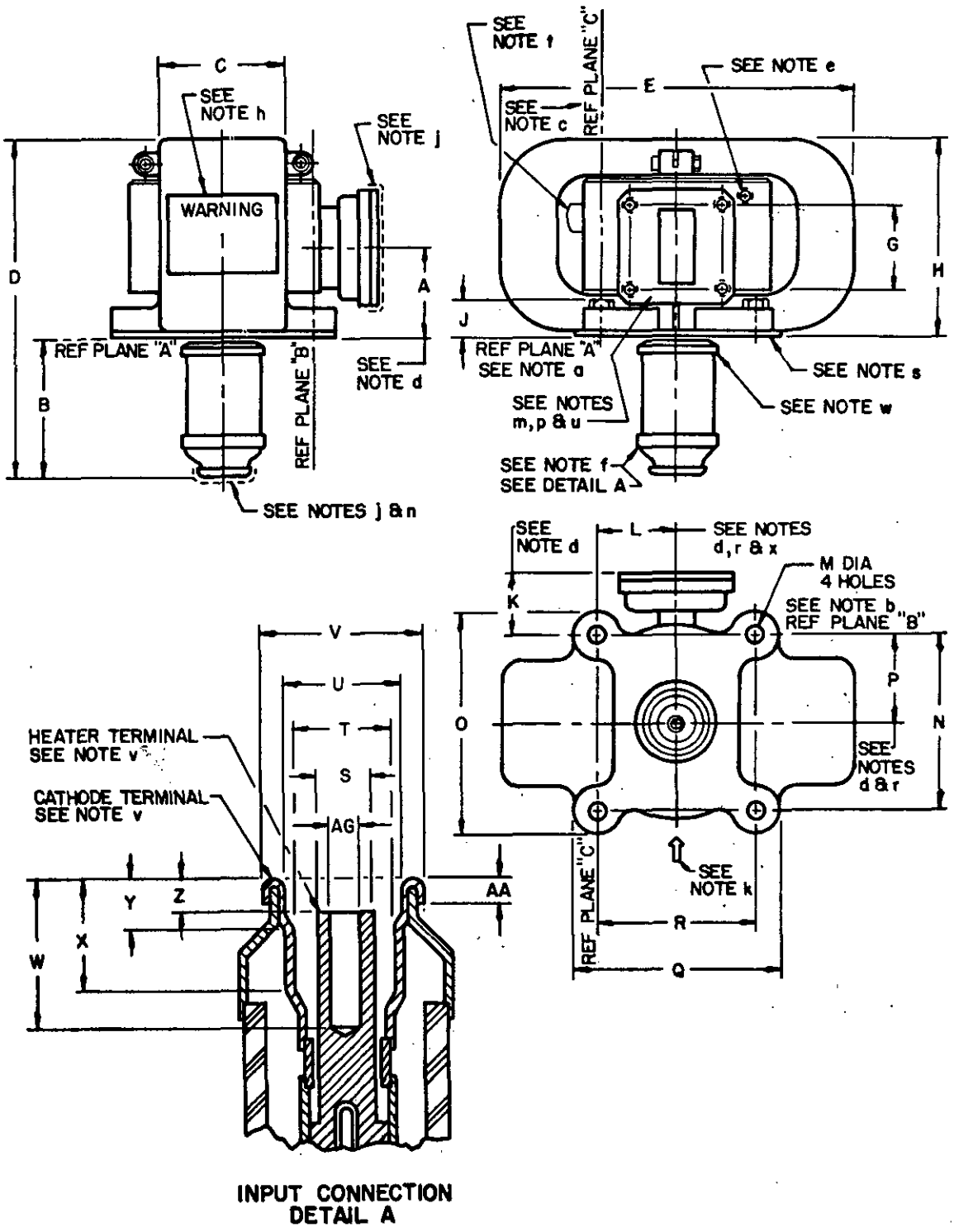
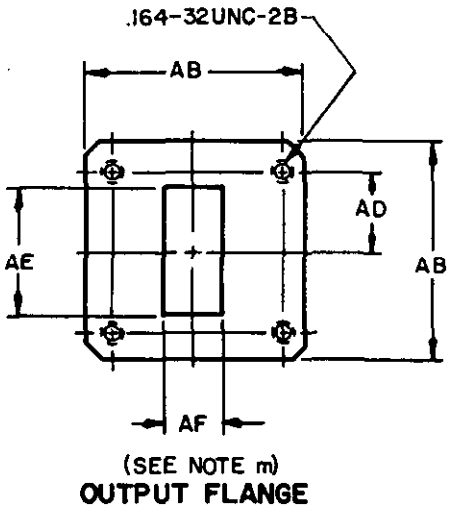


FIGURE 1. Outline drawing of electron tube type DOD-035.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1 See note 16			
A	1.386	1.426	d
B	2.094	2.219	a
F	1.470	1.478	---
G	1.348	1.356	---
J	.596	.614	a
K	.990	1.040	d
M	.276	.286	---
N	2.771	2.791	b
R	2.521	2.541	c
S	.234	.266	---
T	.532	.545	v
U	.610	.650	---
V	.825	.830	---
Z	.125	.187	---
AA	.115	.135	---
AC	.732	.742	---
AD	.671	.681	---
AG	.164	.174	v
Reference dimensions			
C	2.032		---
D	5.375 (136.53)		---
E	5.937 (150.80)		---
H	3.156		---
L	1.265		d, r, x
O	3.625		---
P	1.390		b
Q	3.437		---
W	.750		---
X	.562		---
Y	.156		---
AB	1.830		---
AE	1.122		---
AF	.497		---

FIGURE 1. Outline drawing of electron tube type DOD-035 - Continued.

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NOTES:

a. Reference plane "A" is defined as a plane passing along the face of the mounting plate.
b. Reference plane "B" is defined as a plane perpendicular to plane "A" passing through the axis of the holes, as shown at reference plane "A".
c. Reference plane "C" is defined as a plane mutually perpendicular to planes "A" and "B" passing through the axis of the holes, as shown at reference plane "A".
d. Includes angular as well as lateral deviations.
e. Anode temperature to be measured at this point. The tube serviceability will not be impaired with a screw inserted or removed from this threaded hole.
f. Input bushing temperature to be measured at this point.
g. For vibration of shock testing, the planes of testing shall be reference planes "A" - "B" - "C".
h. Warning: Maintain minimum clearance of 4 inches between magnet and magnetic materials (magnets, steel tools, plates, etc.). Use no magnetic inspection tools.
j. Protective closure. To be removed before tube is used, and attached when the tube is not in use.
k. Indicates direction of body cooling airflow.
m. Mates with modified choke flange, UG-52 B/U (MIL-F-3922/59) (clearance instead of threaded holes).
n. With the tube mounted on the test fixture and with the specified air pressure applied so as to surround the entire input terminal beyond the magnetron mounting plate, the entire magnetron and fixture are to be submerged in water. No bubbles are allowed in a 1-minute interval.
p. With the tube output flange clamped against a modified choke flange (UG-52 B/U) and using a gasket per MS90064-14, and with the specified air pressure applied to the interior of the waveguide submerge entire tube and fixture in water. No bubbles are allowed in a 1-minute interval.
q. A plane passing through the axis of two threaded holes perpendicular to the face of the output flange shall be parallel to planes "A" and "C" within .030.
r. These dimensions apply to the axis of diameter "W".
s. Any portion of the assembly extending below this reference plane shall be within a .625 radius of the axis of the input bushing.
t. Permanently protective closure for the exaust tubulation of the tube shall not restrict airflow nor impair serviceability of the tube.
u. The face of the waveguide flange shall be flat within .010 total indicator reading. The surface of the flange shall be of such quality that a hermetics seal can be effected. (See note p).
v. Heater terminal and cathode terminal shall be concentric within .010.
w. This identifies a north seeking pole. This may be tested by observing the compass needle points to the geographic south. The same needle will point toward the tube's north seeking pole when it is brought near the input stem of the tube. Care should be taken to insure that the tube field does not reverse the polarity of the compass.
x. The axis of the cathode terminal shall be within a radius of .046 of the specified location. (See note d).

FIGURE 1. Outline drawing of electron tube type DOD-035 - Continued.

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